
Online Examination Report

Date:	21.07.2014
Attempts:	1
Examinee:	ATIALUK, Joseph
Date of birth:	12/24/1987
Subject:	030-Flight Performance&Planing
Result:	30 %
Time used:	00:00:13 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	374	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
2	384	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
3	375	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
4	390	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
5	361	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
6	380	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
7	391	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
8	373	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
9	378	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
10	388	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
11	381	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
12	377	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
13	376	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
14	379	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
15	387	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
16	424	32.	PERFORMANCE - AEROPLANES	1,00	1,00
17	118	32.2.2	Effect of Variables on Single-Engine Aeroplane Performance	0,00	1,00
18	160	32.2.5.1	Take-off	0,00	1,00
19	152	32.2.5.4	Landing	0,00	1,00
20	72	32.4.1	Take - off	0,00	1,00
21	123	32.4.1	Take - off	1,00	1,00
22	34	32.4.1.7	Take-off climb	0,00	1,00
23	103	32.4.1.7	Take-off climb	0,00	1,00
24	135	32.4.1.7	Take-off climb	0,00	1,00
25	167	32.4.3	Cruise	1,00	1,00
26	82	32.4.3	Cruise	0,00	1,00
27	216	32.4.3	Cruise	0,00	1,00
28	141	32.4.3.4	Long Range Cruise	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	154	32.4.3.4	Long Range Cruise	0,00	1,00
30	196	32.4.5	Descent	1,00	1,00
31	416	33.	FLIGHT PLANNING AND FLIGHT MONITORING	1,00	1,00
32	92	33.	FLIGHT PLANNING AND FLIGHT MONITORING	0,00	1,00
33	413	33.	FLIGHT PLANNING AND FLIGHT MONITORING	1,00	1,00
34	421	33.	FLIGHT PLANNING AND FLIGHT MONITORING	0,00	1,00
35	418	33.	FLIGHT PLANNING AND FLIGHT MONITORING	0,00	1,00
36	226	33.1.1.1	Routes, airfields, heights and altitudes from VFR charts	0,00	1,00
37	175	33.2.1.2	Courses and distances from en-route charts	0,00	1,00
38	79	33.3.1	General	0,00	1,00
39	252	33.4.1.3	Airway routings and airspace structure	1,00	1,00
40	88	33.5.1.1	Format of Flight Plan	0,00	1,00
Total:30%				12,00	40,00

Attachment 2: List of result by topic

Licence

Date: 18.07.2014

Examinee: ATIALUK, Joseph

Location: Uganda Civil Aviation Authority

Your Result: 12,00 from 40,00 Points (30,00 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	15	5,00	15,00	33,33
32. PERFORMANCE - AEROPLANES	15	4,00	15,00	26,67
32.2. PERFORMANCE CLASS B - SINGLE-ENGINE AEROPLANES	3	0,00	3,00	0,00
32.4. PERFORMANCE CLASS A - AEROPLANES CERTIFICATED UNDER CS 25 ONLY	11	3,00	11,00	27,27
33. FLIGHT PLANNING AND FLIGHT MONITORING	10	3,00	10,00	30,00
33.1. FLIGHT PLANNING FOR VFR FLIGHTS	1	0,00	1,00	0,00
33.2. FLIGHT PLANNING FOR IFR FLIGHTS	1	0,00	1,00	0,00
33.3. FUEL PLANNING	1	0,00	1,00	0,00
33.4. PRE-FLIGHT PREPARATION	1	1,00	1,00	100,00
33.5. ICAO FLIGHT PLAN (ATS Flight Plan)	1	0,00	1,00	0,00
Total	40	12,00	40,00	30,00

1

How much weight could be added at Station 120 without exceeding the aft CG limit?

Aircraft weight 9,500 lb

CG location Station 90.0

Aft CG limit Station 90.5

☒ 110.5 pounds.

☐ 161.0 pounds.

☐ 61.0 pounds.

2

An aircraft is loaded with the CG at the aft limit. What are the performance characteristics compared with the CG at the forward limit?

- ☐ The aft CG provides the lowest stalling speed, the highest cruising speed, and least stability.
- ☐ Cruising speed is lower because of more induced drag created by the elevator or stabilizer being required to provide more lift with an aft CG.
- ☒ The aft CG provides the highest stall speed and cruising speed.

3

What is the maximum weight that could be added at Station 150.0 without exceeding the aft CG limit?

Aircraft weight 5,000 lb
CG location Station 80.0

☐ 70.0 pounds.

☐ 35.9 pounds.

☒ 69.5 pounds.

4

Under which condition is a forward CG most critical?

- ☐ On takeoff.
- ☐ When in an unusual attitude.
- ☒ On landing.

5

What is the maximum allowable weight that may be carried on a pallet which has the dimensions of 24.6 x 68.7 inches?
Floor load limit - 85 lb/sq ft
Pallet weight - 44 lb
Tiedown devices - 29 lb

☐ 953.6 pounds.

☐ 968.6 pounds.

☒ 924.5 pounds.

6

Could 100 pounds of weight be shifted from Station 130.0 to Station 30.0 without exceeding the forward CG limit?

Total weight 2,800 lb
CG location Station 120.0

☐ No; the new CG would be located at Station 116.89.

☒ Yes; the new CG would be located at Station 117.89.

☐ No; the new CG would be located at Station 116.42.

7

As the CG location is changed, recovery from a stall becomes progressively

☒ **more difficult as the CG moves either forward or rearward.**

☐ **more difficult as the CG moves rearward.**

☐ **less difficult as the CG moves rearward.**

8

How much weight could be added at Station 160 without exceeding the aft CG limit?

Aircraft weight 8,300 lb
CG location Station 90.0
Aft CG limit Station 90.5

☒ 16.5 pounds.

☐ 59.7 pounds.

☐ 13.9 pounds.

9

What is the location of the CG if 60 pounds are removed from Station 70?

Aircraft weight 8,420 lb

CG location Station 85

☐ **84.9.**

☒ **85.1.**

☐ **84.1.**

10

What is characteristic of the indicated airspeed if the CG is at the most forward allowable position and constant power and altitude are maintained?

- ☐ Indicated airspeed will be less than it would be with the CG in the most rearward allowable position.
- ☐ There is no relationship between CG location and indicated airspeed.
- ☒ Indicated airspeed will be greater than it would be with the CG in the most rearward allowable position.

11

Could 100 pounds of weight be shifted from Station 30.0 to Station 120.0 without exceeding the aft CG limit?

Total weight 4,750 lb

CG location Station 115.8

☒ Yes; the new CG would be located at Station 117.69.

☐ No; the new CG would be located at Station 118.15.

☐ Yes; the CG would remain at Station 115.8.

12

What is the location of the CG if 146 pounds are removed from Station 150?

Aircraft weight 7,152 lb

CG location Station 82

☐ 83.4.

☐ 80.6.

☒ 81.3.

13

What is the location of the CG if 90 pounds are removed from Station 140?

Aircraft weight 6,230 lb

CG location Station 79

☒ **77.9.**

☐ **78.1.**

☐ **79.9.**

14

How much weight must be shifted from Station 150.0 to Station 30.0 to move the CG to exactly the aft CG limit?

Total weight 7,500 lb

CG location Station 80.5

☐ 65.8 pounds.

☒ 62.5 pounds.

☐ 68.9 pounds.

15

As the CG moves aft, an aircraft becomes

- ☐ less stable and less controllable.
- ☒ more stable and controllable as long as the aft CG is not exceeded.
- ☐ less stable, yet easier to control.

16

To level off from a descent maintaining the descending airspeed, the pilot should lead the desired altitude by approximately

☐ 60 feet.

☒ 50 feet.

☐ 20 feet.

17

Which statement is correct for a descent without engine thrust at maximum lift to drag ratio speed?

- ☐ The mass of an aeroplane does not have any effect on the speed for descent.
- ☐ The higher the gross mass the greater is the speed for descent.
- ☐ The higher the average temperature (OAT) the lower is the speed for descent.
- ☒ The higher the gross mass the lower is the speed for descent.

18

The speed V_1 is defined as

- ☐ speed for best angle of climb.
- ☒ take-off decision speed.
- ☐ take-off climb speed.
- ☒ engine failure speed.

19

Which of the following factors will lead to an increase of ground distance during a glide, while maintaining the appropriate minimum glide angle speed?

- ☒ Headwind.
- ☐ Increase of aircraft mass.
- ☐ Decrease of aircraft mass.
- ☐ Tailwind.

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Subject: 030-Flight Performance&Planing
Date: 2014-07-18

20

During certification flight testing on a four engine turbojet aeroplane the actual take-off distances measured are: - 3050 m with failure of the critical engine recognised at V1 - 2555 m with all engines operating and all other things being equal The take-off distance adopted for the certification file is:

☐ 3513 m

☐ 2938 m

☒ 2555 m

☐ 3050 m

21

What will be the effect on an aircraft's performance if aerodrome pressure altitude is decreased?

- ☐ It will increase the take-off distance required.
- ☐ It will increase the accelerate stop distance.
- ☒ It will decrease the take-off distance required.
- ☐ It will increase the take-off ground run.

22

Which of the following statements is correct?

- ☒ The take-off distance with one engine out is independant of the wind component.
- ☐ The accelerate stop distance required is independant of the runway condition.
- ☐ The performance limited take-off mass is independant of the wind component.
- ☐ The climb limited take-off mass is independant of the wind component.

23

What happens to the drag of a jet aeroplane if, during the initial climb after take off, constant IAS is maintained?
(Assume a constant mass.)

- ☐ The drag increases considerably.
- ☒ The drag decreases.
- ☐ The drag remains almost constant.
- ☐ The drag increases initially and decreases thereafter.

24

With an true airspeed of 194 kt and a vertical speed of 1 000 ft/min, the climb gradient is about :

☐ 0.08

☐ 3°

☒ 5°

☐ 0.03

25

Density altitude is the

- ☐ altitude read directly from the altimeter
- ☒ pressure altitude corrected for 'non standard' temperature
- ☐ altitude reference to the standard datum plane
- ☐ height above the surface

26

Which of the following statements is correct?

- ☐ Induced drag increases with increasing speed.
- ☐ Induced drag decreases with increasing angle of attack.
- ☒ Induced drag is independant of the speed.
- ☐ Induced drag decreases with increasing speed.

27

Which force compensates the weight in unaccelerated straight and level flight ?

☒ the resultant from lift and drag

☐ the lift

☐ the drag

☐ the thrust

28

What effect has a tailwind on the maximum endurance speed?

- ☐ The IAS will be decreased.
- ☐ No affect
- ☒ The IAS will be increased.
- ☐ Tailwind only effects holding speed.

29

A lower airspeed at constant mass and altitude requires

- ☒ a higher coefficient of lift.
- ☐ less thrust and a lower coefficient of lift.
- ☐ more thrust and a lower coefficient of drag.
- ☒ more thrust and a lower coefficient of lift.

30

A jet aeroplane descends with constant Mach number. Which of the following speed limits is most likely to be exceeded first?

- ☒ Maximum Operating Speed
- ☐ High Speed Buffet Limit
- ☐ Maximum Operational Mach Number
- ☐ Never Exceed Speed

31

If an aircraft is consuming 91 pounds of fuel per hour and groundspeed is 168 knots, how much fuel is required to travel 457 NM?

☒ 248 pounds.

☐ 265 pounds.

☐ 291 pounds.

32

From which of the following would you expect to find details of the Search and Rescue organisation and procedures (SAR) ?

- ☒ ATCC broadcasts
- ☐ SIGMET
- ☐ NOTAM
- ☐ AIP (Aeronautical Information Publication)

33

GIVEN:

Usable fuel at takeoff 40 gal, Fuel consumption rate 12.2 gal/hr Constant
groundspeed 120 kts
Flight time since takeoff 1 hr 30 min

☐ 156 NM.

☐ 216 NM.

☒ 121 NM.

34

How is your flight plan closed when your destination airport has IFR conditions and there is no control tower or Flight control Center(FCC) on the field?

- ☐ Upon landing, you must close your flight plan by radio or by telephone to any FCC or ATC facility.
- ☐ You may close your flight plan any time after starting the approach by contacting any FCC or ATC facility.
- ☒ The ARTCC controller will close your flight plan when you report the runway in sight.

35

If an aircraft has a transponder, encoding altimeter, and DME, the proper equipment suffix to be entered on a flight plan is

☒ R.

☐ A.

☐ U.

36

Given : true track 017; W/V 340/30; TAS 420 kt Find : wind correction angle (WCA) and ground speed (GS)

- ☐ WCA -2° ; GS 396 kt
- ☒ WCA +2° ; GS 396 kt
- ☐ WCA +2° ; GS 416 kt
- ☐ WCA -2° ; GS 426 kt

37

You are flying a constant compass heading of 252° . Variation is 22°E , deviation is 3°W and your INS is showing a drift of 9° right. True track is ?

☐ 280°

☐ 242°

☐ 224°

☒ 262°

38

The fuel burn off is 200 kg/h with a relative fuel density of 0,8. If the relative density is 0.75 the fuel burn will be:

☒ 213 kg/h

☐ 267 kg/h

☐ 200 kg/h

☐ 188 kg/h

39

From which of the following would you expect to find the dates and times when temporary danger areas are active

- ☐ *Only AIP*
- ☐ *RAD/NAV charts*
- ☒ *NOTAM and AIP*
- ☐ *SIGMET*

40

A "current flight plan" is a:

- ☐ *flight plan in the course of which radio communication should be practised between aeroplane and ATC.*
- ☐ *filed flight plan with amendments and clearance included.*
- ☐ *flight plan with the correct time of departure.*
- ☒ *filed flight plan.*